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CENTRE NATIONAL D'ÉTUDES SPATIALES

XXIXth Telemac User Conference
Front type observation for remote sensing data assimilation for flood modelling

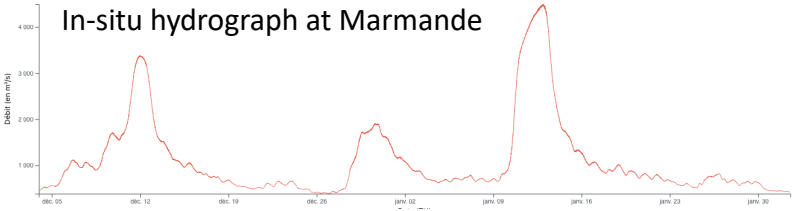
Quentin Bonassies (CECI, CERFACS / UMR 5318 CNRS)

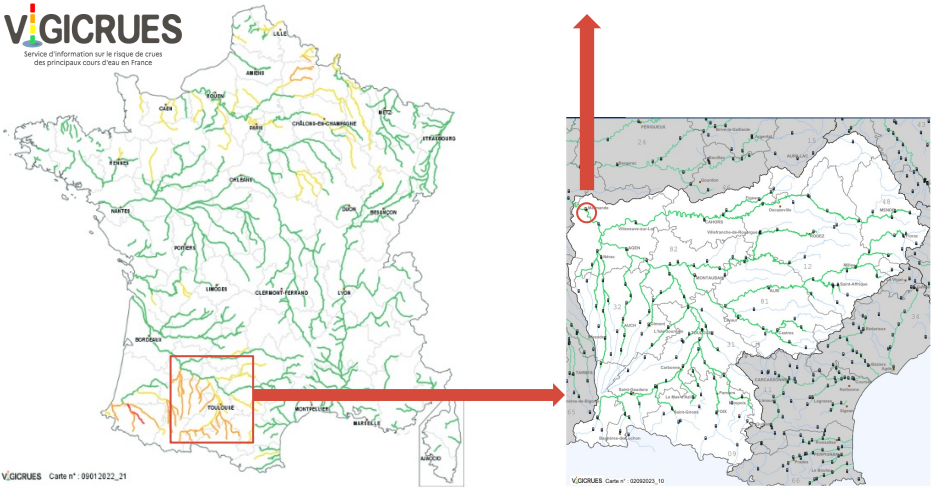
11th – 13th Oct. 2023

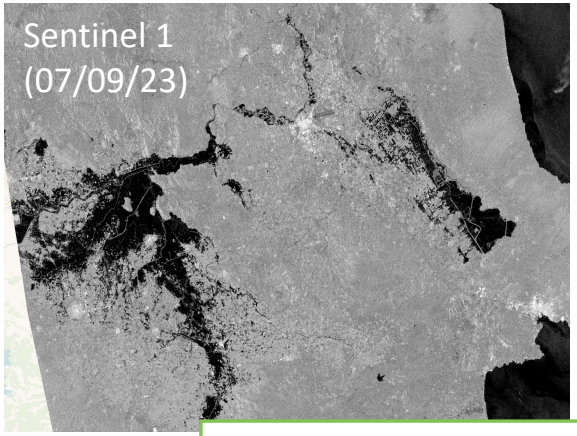
Sophie Ricci, T.H. Nguyen, A. Piacentini, M. Sadki (CECI, CERFACS / UMR 5318 CNRS)
Christophe Fatras, A. Andral (CLS),
Santiago Peña-Luque, R. Rodriguez Suquet (CNES)

Flood monitoring and flood risk management

Flood Monitoring in France

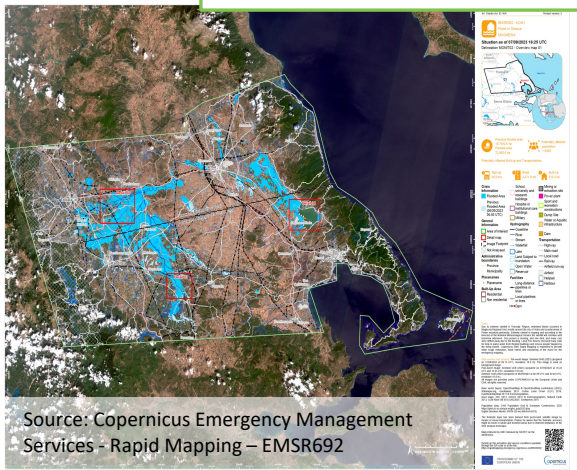






SAR image
During event

Recent event in Greece



Copernicus
Emergency
Services
Rapid Mapping

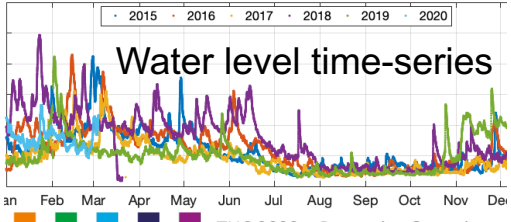
Data driven modelling

Making the most of in-situ and remote sensing data to estimate river discharge and forecast the floodplain dynamics

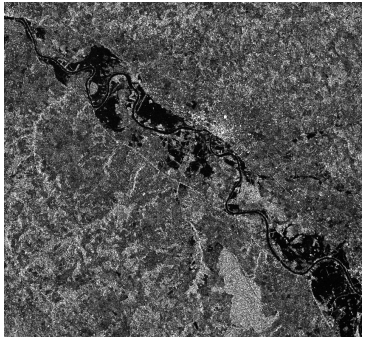
VigiCrue in-situ network



Marmande

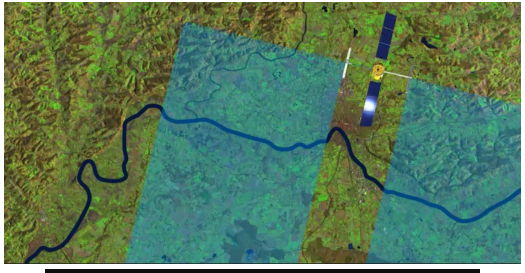


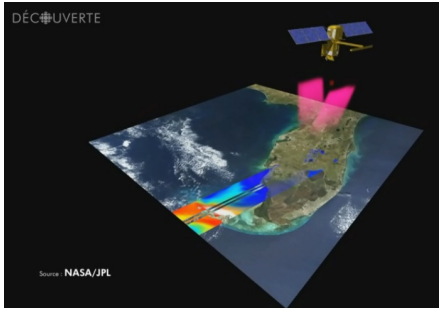
Sentinel 1&2 + FloodML



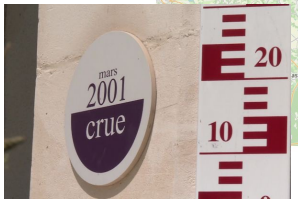


SWOT









High water marks

Flood risk and river hydrodynamics modeling with Telemac-2D

- Uncertainties in data,
- Uncertainties in human expertise
- Uncertainties related to the nature of the events
- …

- Uncertainties related to observations
- Uncertainties related to numerical models for meteorological, hydrological and hydraulic forecasting



Flood at Marmande (FR)
Source: France 2 (Feb. 04th 2021)

Shallow-Water Equation (2D formulation)

Advection

Grad. of hydrostatic pressure

Source term, friction

Diffusion, Turbulence, Dispersion

Continuity

$$\frac{\partial h}{\partial t} + \frac{\partial}{\partial x}(hu) + \frac{\partial}{\partial y}(hv) = 0$$

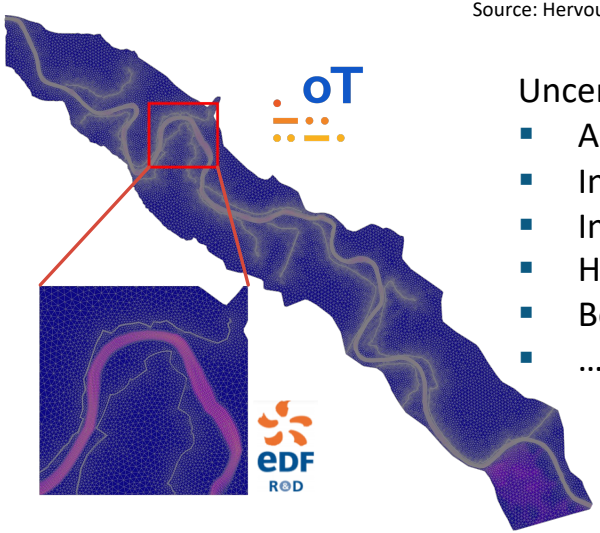
Momentum along x

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -g\frac{\partial z}{\partial x} + F_x + \frac{1}{h} div\left(hv_e \overrightarrow{grad}(u)\right)$$

Momentum along y

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -g\frac{\partial z}{\partial y} + F_y + \frac{1}{h} div\left(hv_e \overrightarrow{grad}(v)\right)$$

Source: Hervouet, 2007



Uncertainties due to:

- Assumption of the model
- Initial bathymetry
- Initial conditions
- Hydraulic forcing
- Boundary conditions
- …

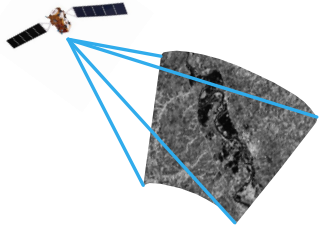
Data Assimilation (DA) for flood forecasting

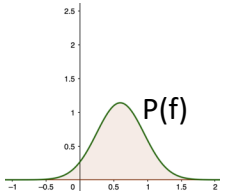
"True" flood extent

Measurement errors

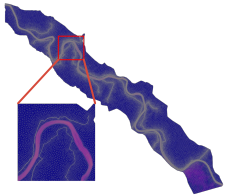
ε

EO-based flood observations



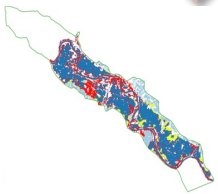


Uncertain forcings/parameters

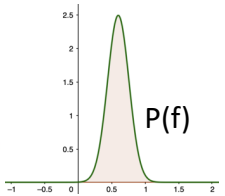


Flood inundation models

Data Assimilation



$CSI = 68.96\%$



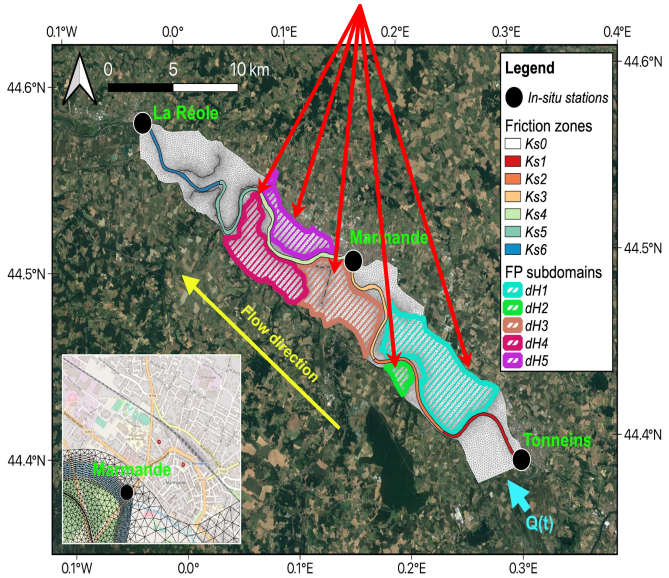
Updated model states

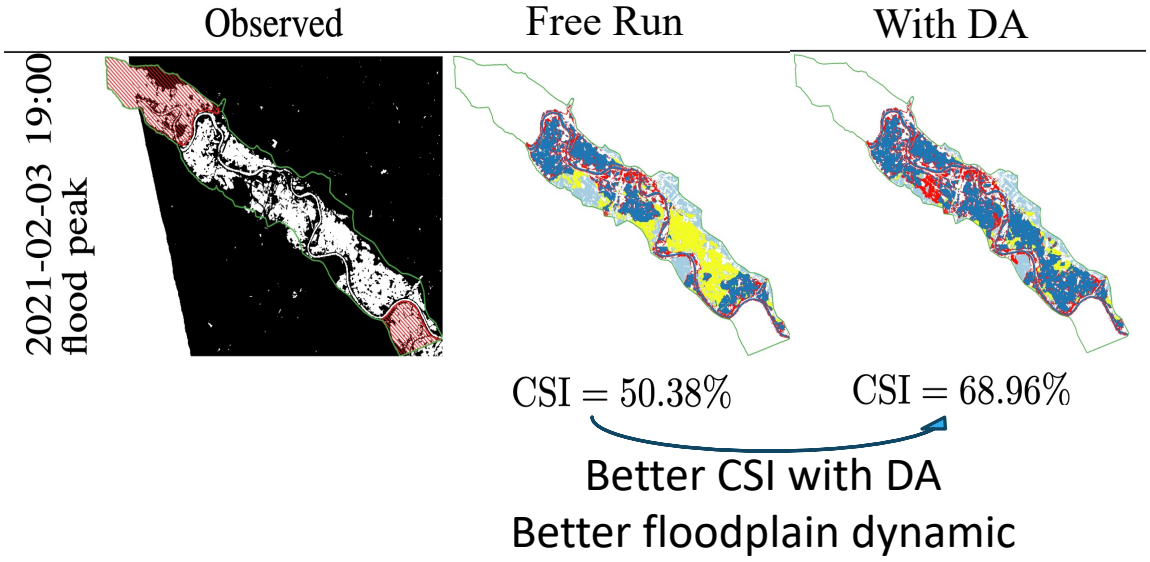
Data assimilation of remote sensing data for flood extent

Assimilation of Wet Surface Ratio (WSR):
ratio of wet pixels within an area

5 areas = 5 WSR computed from SAR-derived flood extents

Control vector for data assimilation:
7 Ks within riverbed
1 inflow multiplicative parameter
5 floodplain corrective parameters

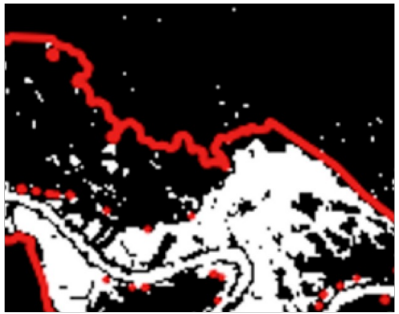


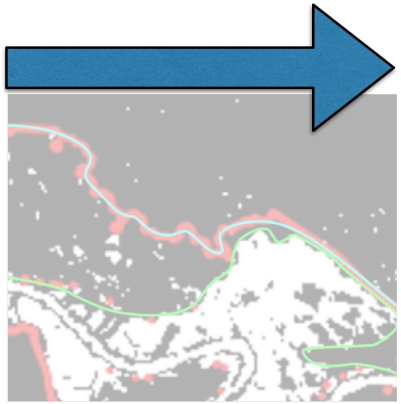


Source: T.H. Nguyen *et al.,* 2022 (WRR) ; T.H. Nguyen *et al.,* 2023 (under review)

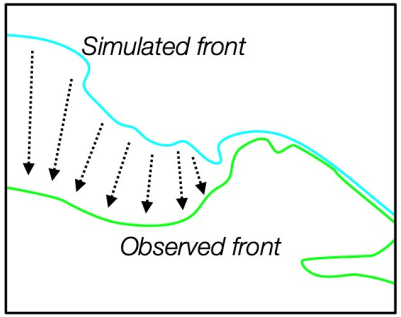
Front type observation for remote sensing data assimilation

From pixel analysis…





…To a flood front analysis

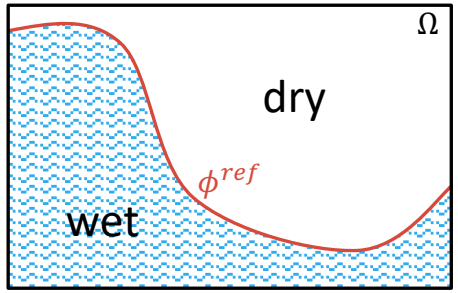


- Red: Simulated free run

- White: Wet pixel
- Black: Dry pixel

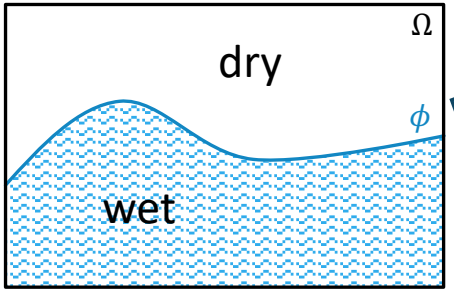
Level-set approach for Data Assimilation
The Chan-Vese contour fitting metric

Image of reference



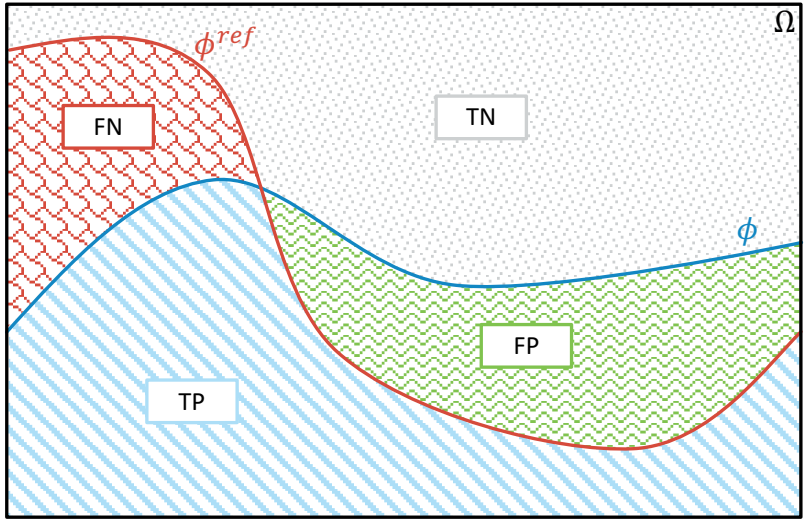
ϕ^{ref} = Contour of reference

Simulated image



ϕ = Simulated contour

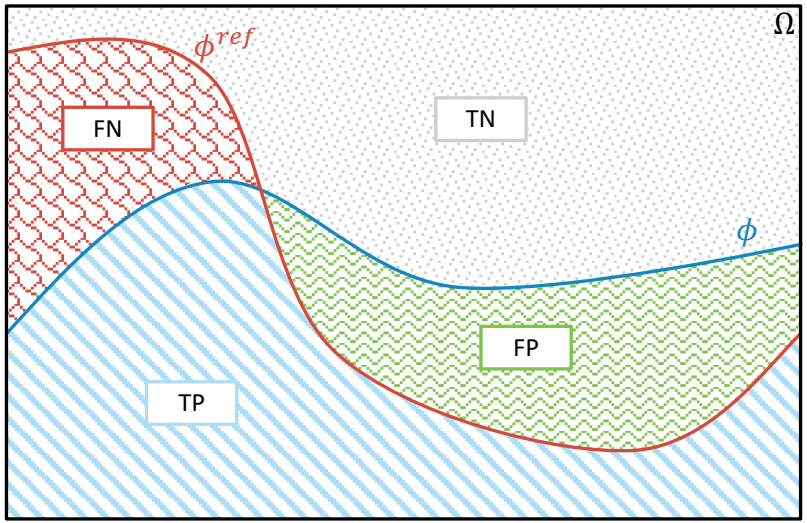
Contingency map



TN: True Negative
FN: False Negative

TP: True Positive
FP: False Positive

Level-set approach for Data Assimilation
The Chan-Vese contour fitting metric



C_1 and C_0, two scalars, are defined as follow:

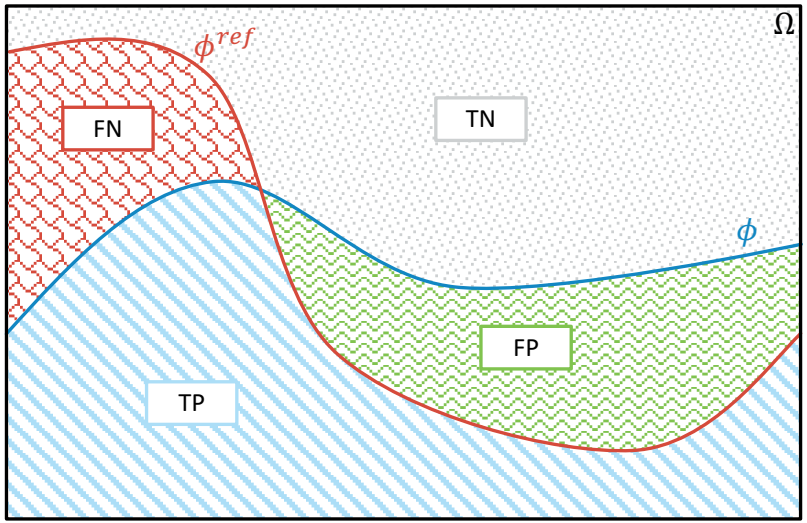
$$C_0 \approx \frac{FN}{TN + FN} \; and \; C_1 \approx \frac{TP}{TP + FP}$$

TN: True Negative
TP: True Positive
FN: False Negative
FP: False Positive

C_0 represents the ratio of misdetected unflooded pixels to the total number of unflooded pixels.

C_1 represents the ratio of well-detected flooded pixels to the total number of flooded pixels.

Level-set approach for Data Assimilation
The Chan-Vese contour fitting metric



C_1 and C_0, two scalars, are defined as follow:

$$C_0 \approx \frac{FN}{TN + FN} \; and \; C_1 \approx \frac{TP}{TP + FP}$$

TN: True Negative
TP: True Positive
FN: False Negative
FP: False Positive

Chan-Vese contour fitting distance:

$$D(\phi^{\mathrm{ref}}, \phi) = D^{+}(\phi^{\mathrm{ref}}, \phi) + D^{-}(\phi^{\mathrm{ref}}, \phi)$$

Where:

$$D^{+}(\phi^{\mathrm{ref}}, \phi) = Hv(\phi)\left(\phi^{\mathrm{ref}} - Cmax(\phi^{\mathrm{ref}}, \phi)\right)$$

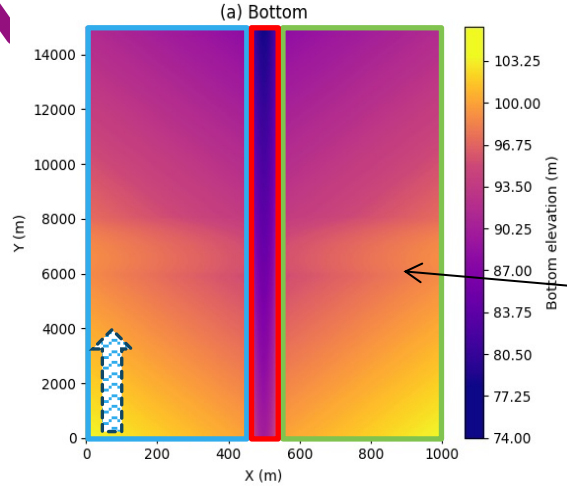
$$D^{-}(\phi^{\mathrm{ref}}, \phi) = (1 - Hv(\phi))\left(\phi^{\mathrm{ref}} - Cmin(\phi^{\mathrm{ref}}, \phi)\right)$$

With:

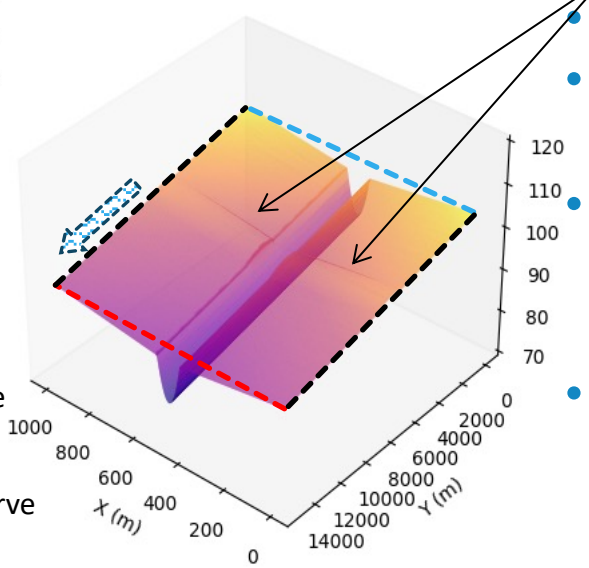
$$Cmax(\phi^{\mathrm{ref}}, \phi) = max(C_0; C_1)$$
$$Cmin(\phi^{\mathrm{ref}}, \phi) = min(C_0; C_1)$$

Sources: Chan and Vese (2001); Rochoux et al. (2018); Zhang et al. (2019); Zhang, PhD Manuscript (2018)

The Valley test case



Ks = 60 | Ks = 35 | Ks = 15



Prescribed flowrate

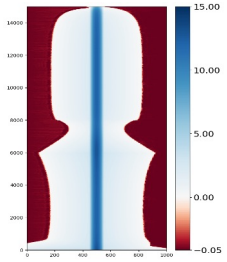
Wall

Stage-discharge curve

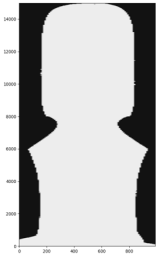
- Mean node density: 12.5 m
- Domain size:
  - Length = 15 km
  - Width = 1 km
- Around 115,000 nodes
- 2 extrusions in the floodplains
- Constant slope in x-axis: 0.8%
- Uniform slope in floodplains: 8%
- Three Strickler coefficients
  - One for each floodplain (Ks = 15 and Ks = 60)
  - One for riverbed (Ks = 35)
- Two liquid boundaries
  - Upstream: Prescribed flowrate
  - Downstream: Stage-Discharge curve

Data Assimilation for flood forecasting in the Valley test case

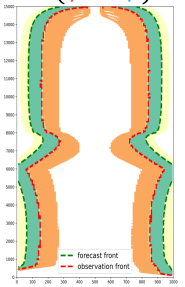
H – 0.05 (m)

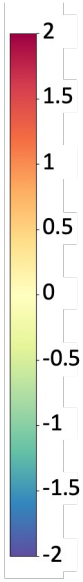


Synthetic EO-based Flood observations



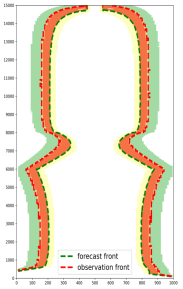
$D^+(\phi^{ref}, \phi)$

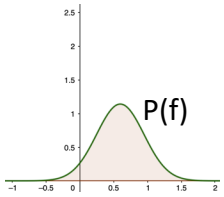




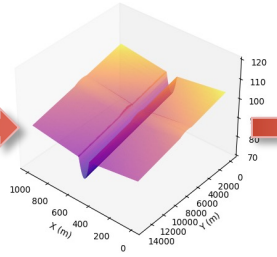
Using Chan-Vese Distance

$D^-(\phi^{ref}, \phi)$

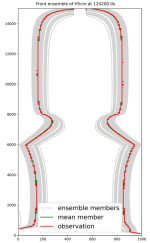




Uncertain forcings/parameters

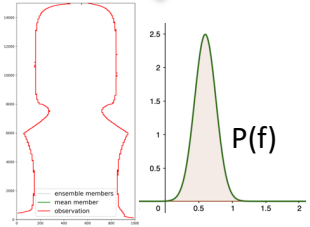


Flood inundation models



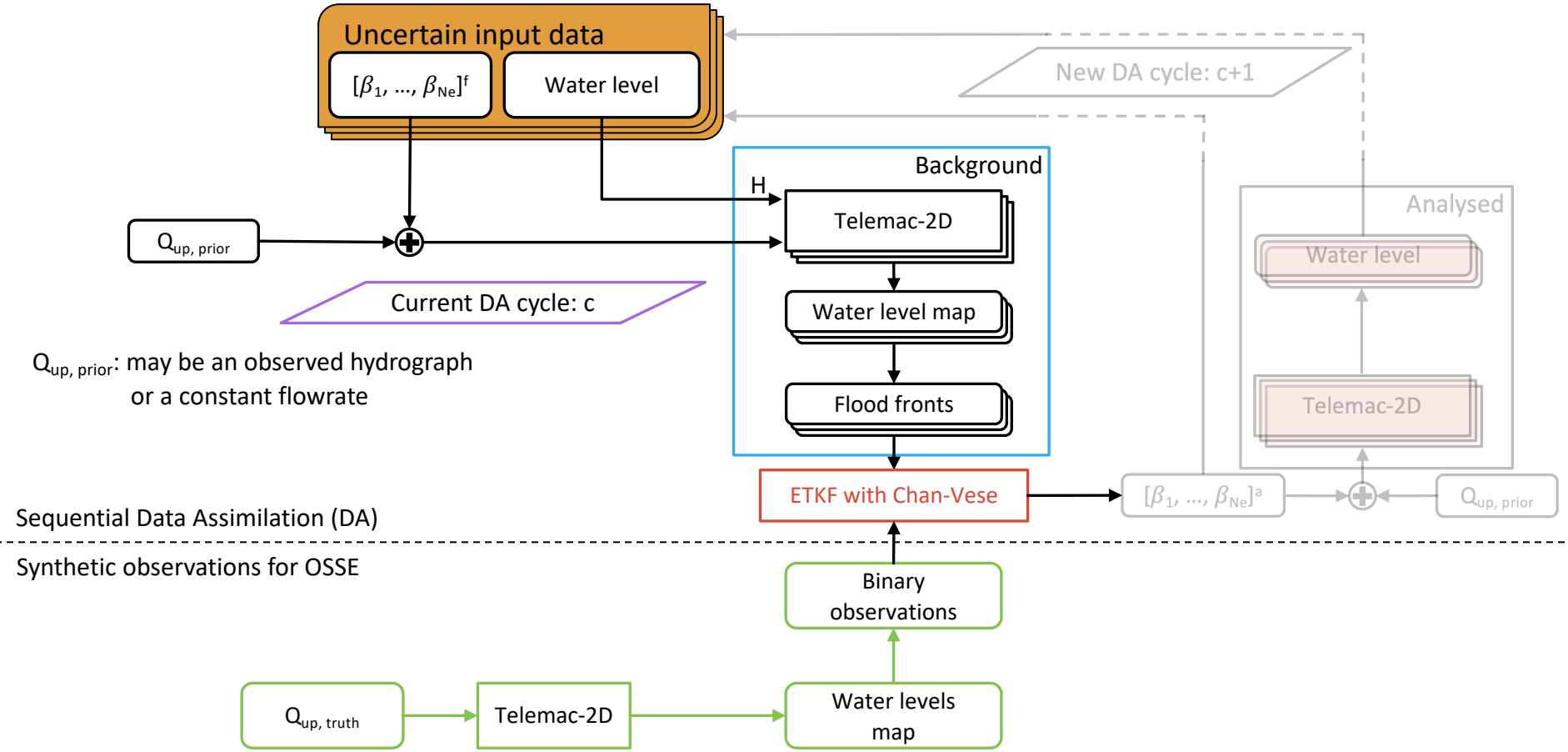
"Open loop" model states

Data Assimilation

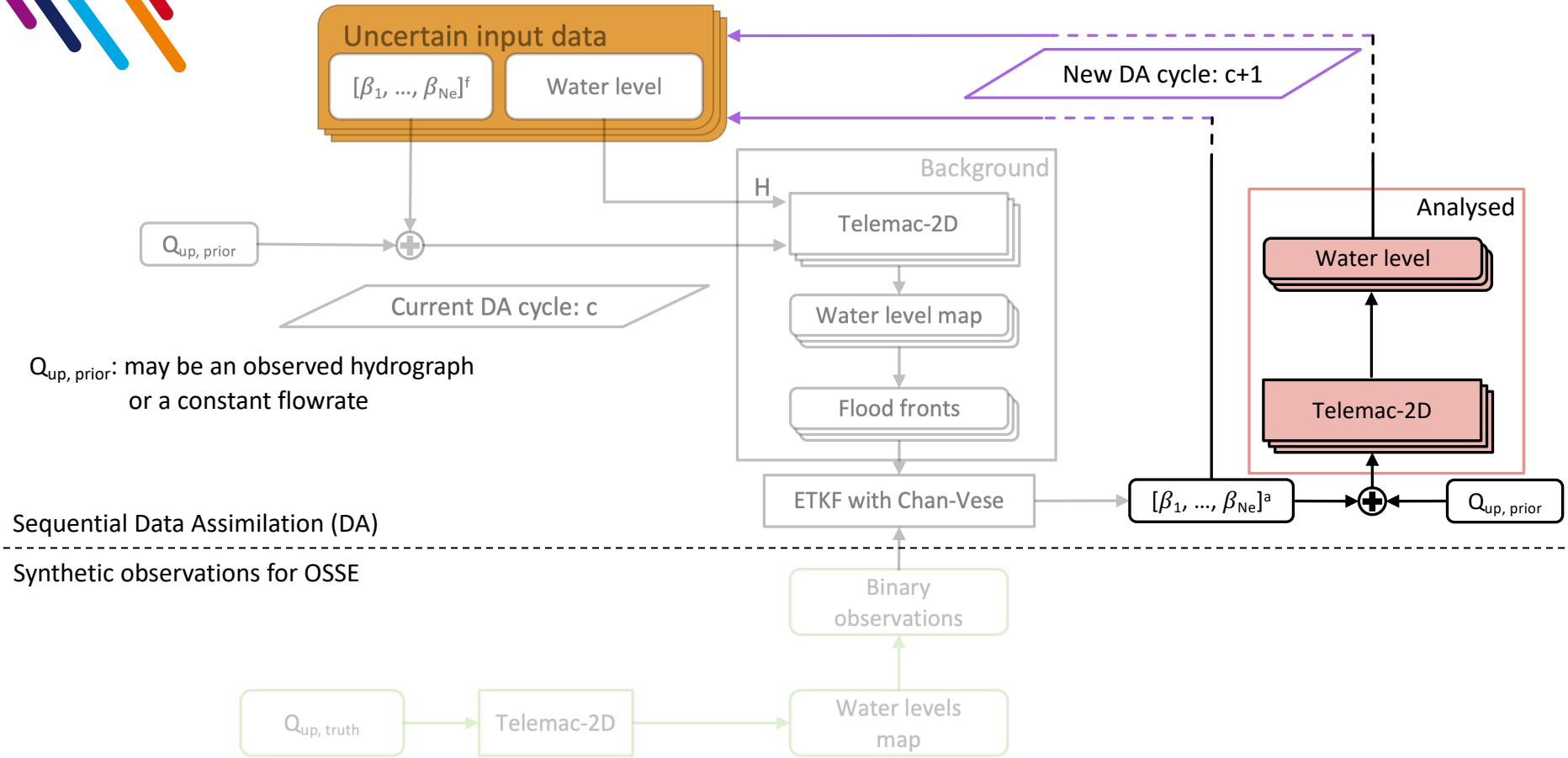


Updated model states and parameters

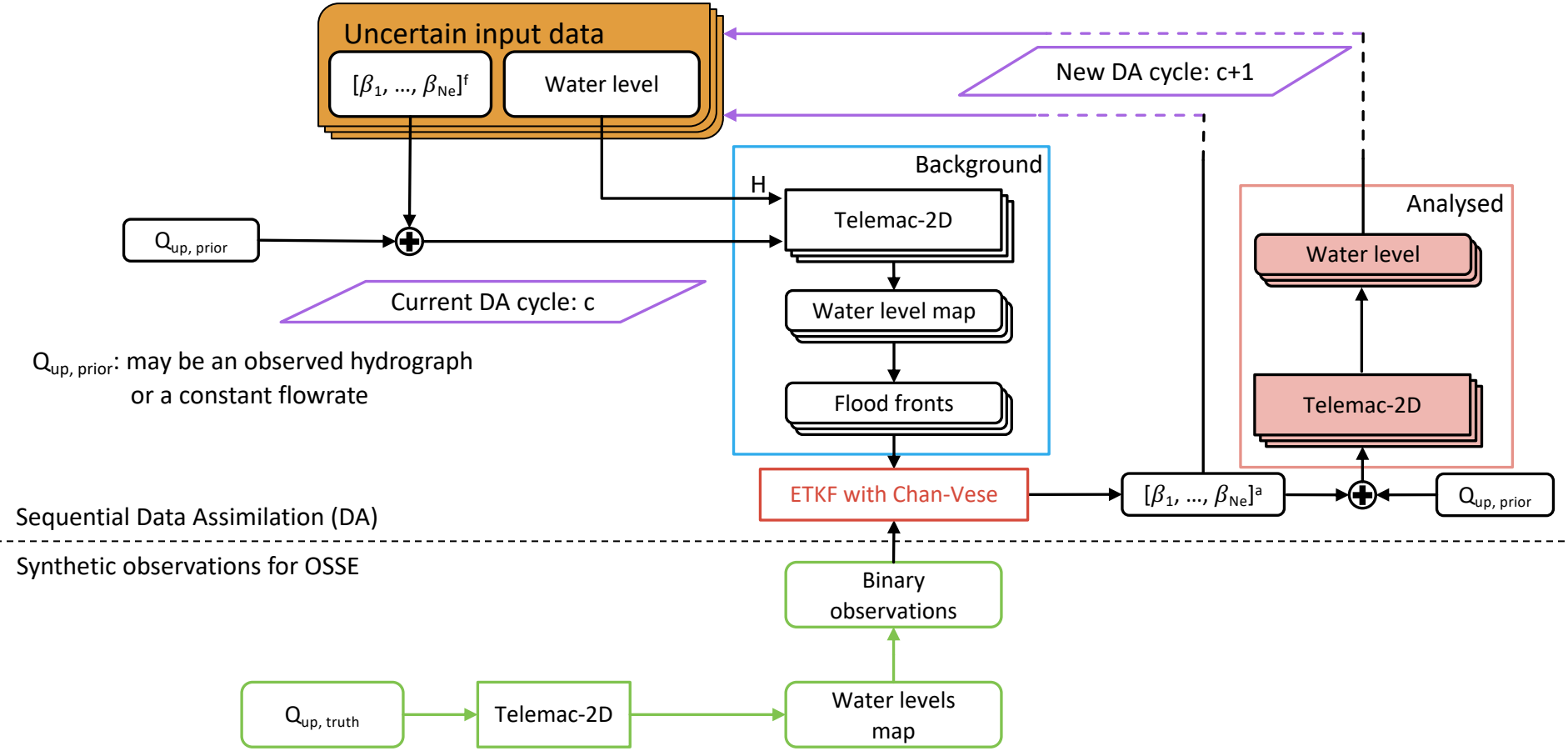
The ETKF algorithm and Chan-Vese



The ETKF algorithm and Chan-Vese



The ETKF algorithm and Chan-Vese



The observing system simulation experiment (OSSE) framework

Ensemble size per experiment:

60 members

Truth setting:

$Q_{up,\ truth} = 5000\ m^3/s$

Control vector:

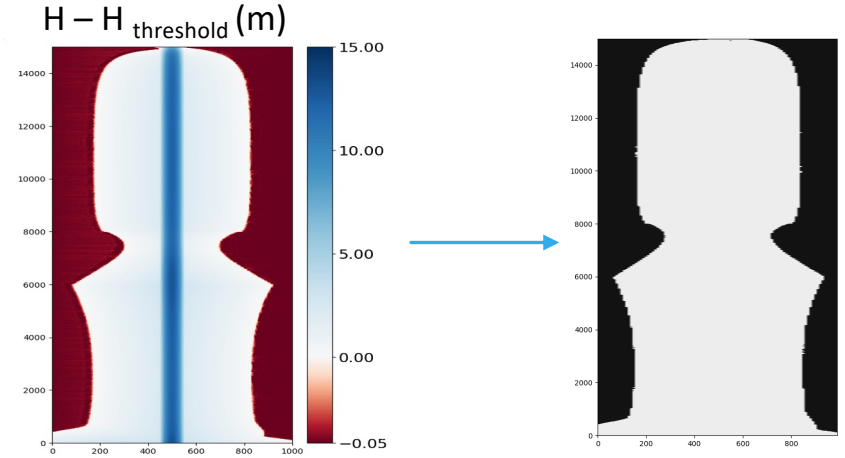
$x = [\beta]$

with $\beta \sim \mathcal{N}(0, 700^2)$

	a prior settings
Exp. A	$Q_{up,\ prior} = 4500\ m^3/s$
Exp. B	$Q_{up,\ prior} = 5500\ m^3/s$

Synthetical-water mask

$H_{threshold} = 0.05$ m



Constant water flowrate

12 hours simulated

Running time: around 2 min on 6 nodes of Intel Xeon Gold 6140 core (CERFACS HPC)

Results of Exp. A ($Q_{up,\ prior}$ = 4500 m^3/s)

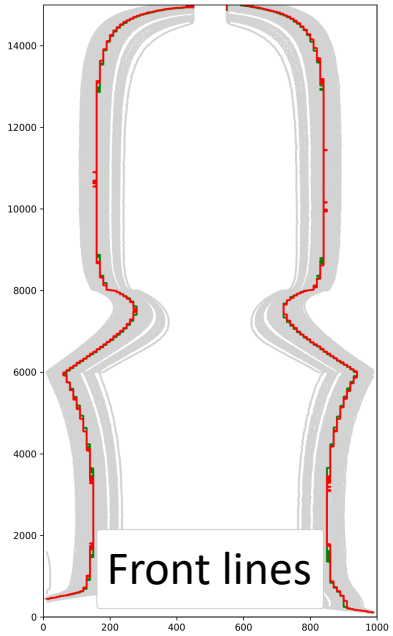
- $Q_{up,\ prior}$ = 4500 m^3/s
- $Q_{up,\ truth}$ = 5000 m^3/s
  - β_{seek} = 500 m^3/s (additional factor)

- Four assimilation cycles
- Three images analysed per cycle
  - β converges in 2 cycles for exp. A

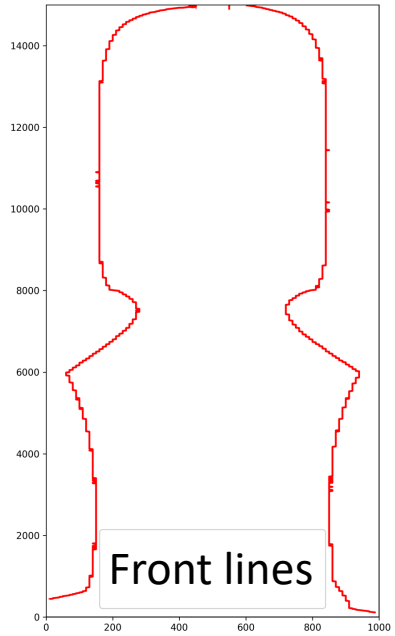
- Time to complete experiment (60 members)
  - ≈ 75 minutes
  - With 360 processors used in total
    And 6 processors per run (member)

On the last DA cycle

Forecast step

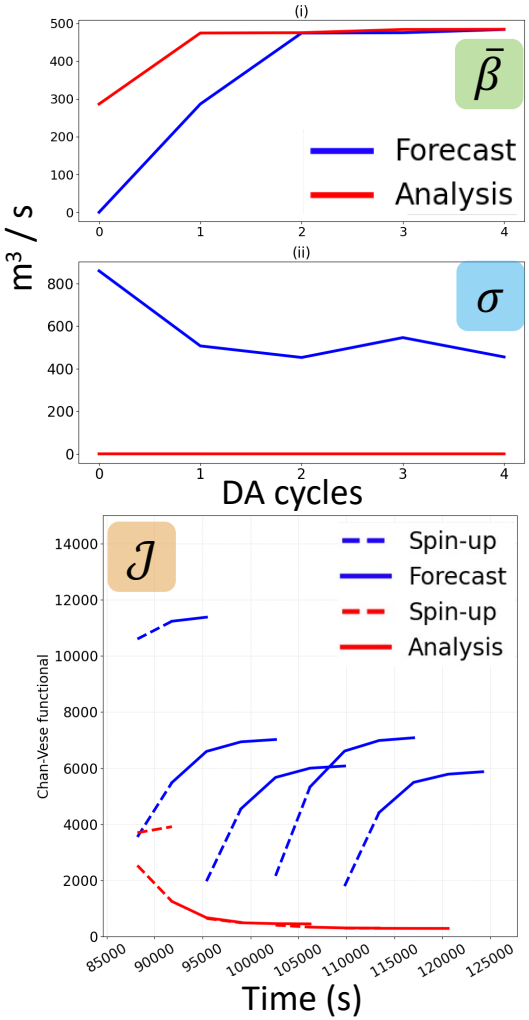


Analysis step



ensemble members
mean member
observation

Results of Exp. A ($Q_{up,\ prior}$ = 4500 m^3/s)



$\bar{\beta}$: ensemble mean of β

$$\bar{\beta} = \frac{1}{60}\sum_{i=1}^{60}\beta_i$$

σ : ensemble standard deviation (std) of β

$$\sigma = \sqrt{\frac{\sum_{i=1}^{60}(\beta_i - \bar{\beta})^2}{60}}$$

$\mathcal{J}$: mean Chan-Vese functional

As an analytical metric (similarity score between two images)

$$\mathcal{J} = \mathcal{J}^+ + \mathcal{J}^-$$

With:

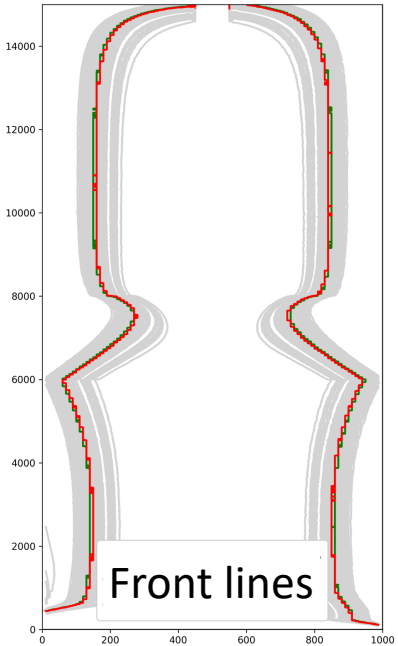
$$\mathcal{J}^+(\phi^{\mathrm{ref}}, \phi) = \int_\Omega Hv(\phi)\left(\phi^{\mathrm{ref}} - Cmax(\phi^{\mathrm{ref}}, \phi)\right)^2 dxdy$$

$$\mathcal{J}^-(\phi^{\mathrm{ref}}, \phi) = \int_\Omega (1 - Hv(\phi))\left(\phi^{\mathrm{ref}} - Cmin(\phi^{\mathrm{ref}}, \phi)\right)^2 dxdy$$

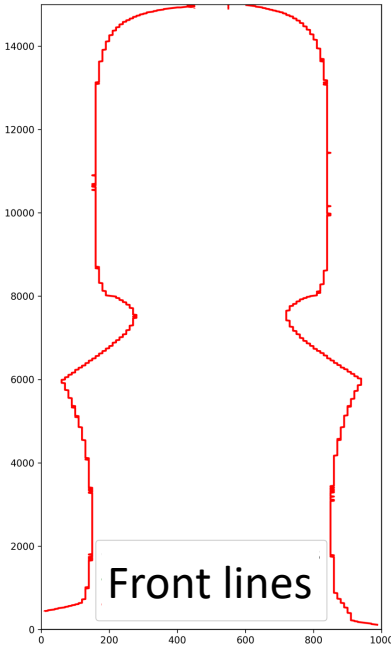
Results of Exp. B ($Q_{up,\ prior}$ = 5500 m³/s)

On the last DA cycle

Forecast step



Analysis step



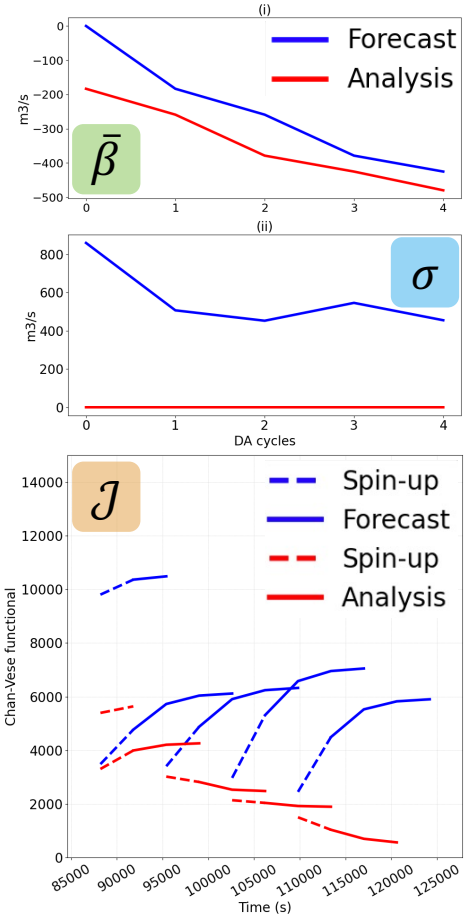
ensemble members
mean member
observation

$\bar{\beta}$: ensemble mean of β

σ: Ensemble std of β

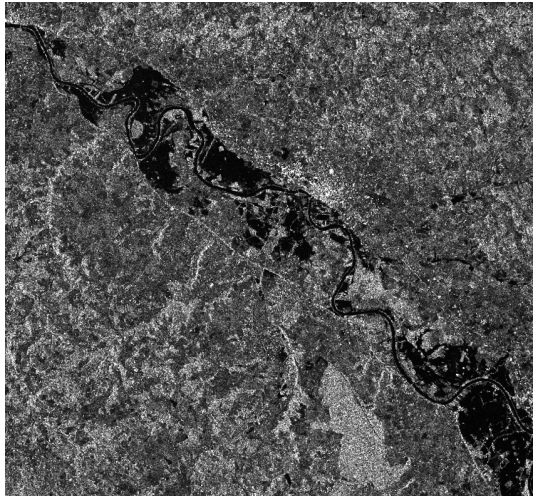
$\mathcal{J}$: Mean Chan-Vese functional

As an analytical metric (similarity score between two images)



Perspectives on Chan Vese-ETKF work

- New experiment with an observation created with a different topography:
  - Observation: with a certain topography (*e. g.* extrusion in floodplains)
  - Forecast ensemble with another topography (*e. g.* floodplains with constant slope)
- Add spatial control vector:
  - Modification of the topography spatially with the DA analysis;
  - Modification of the friction coefficient spatially with the DA analysis



- Test with a more realistic synthetic SAR observation:
  - Add random noise;
  - Images not in the complete domain…
- Test on the Garonne catchment model on 2019/2021 flood events:
  - Not a toy model
  - Real observations

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THANK YOU

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Appendix

Flood monitoring and flood risk management

Occurence of Disasters

Source: CRED (2023), 2022: Disasters in numbers

Figure 1

Number of disasters by continent and top 10 countries[8] in 2022

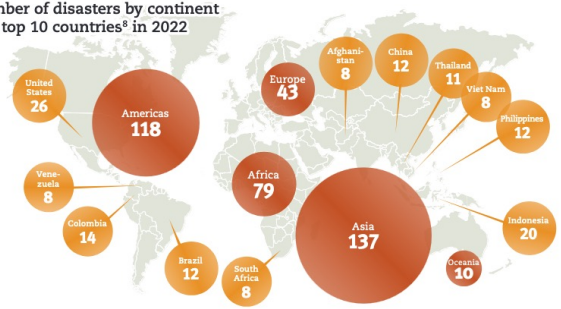
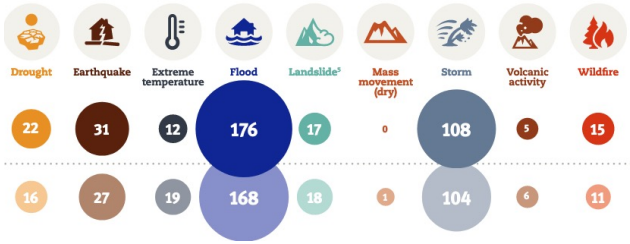


Figure 2

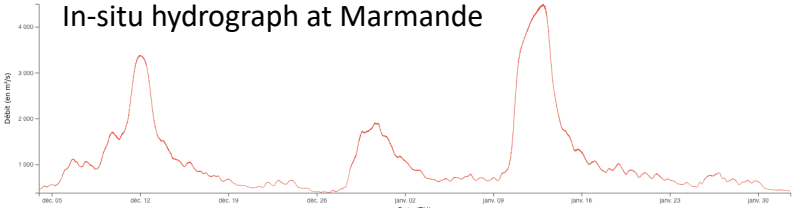
Occurrence by disaster type: 2022 compared to the 2002-2021 annual average

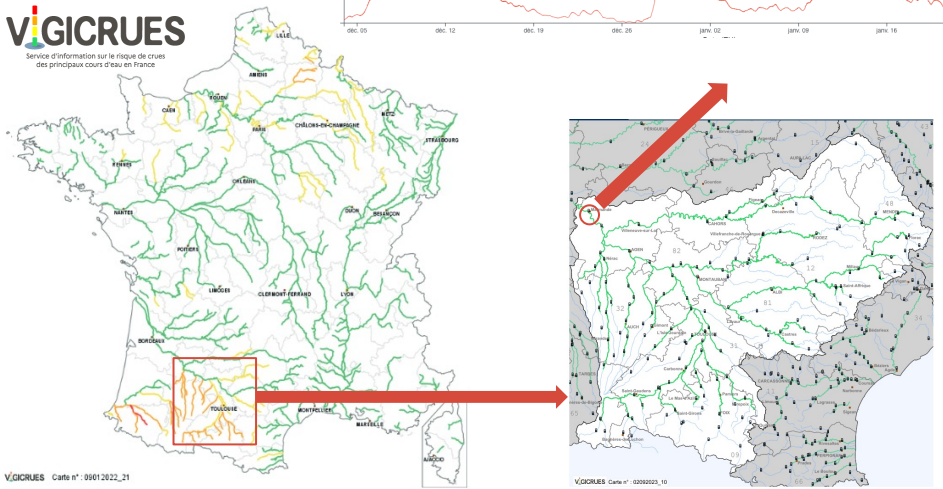
370 2002 to 2021 < 387 in 2022



Occurrence of floods > other disasters

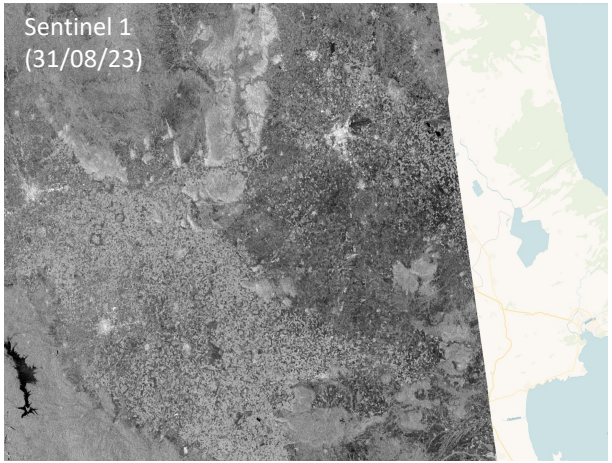
176 floods events > 168 floods





Recent Event in Greece

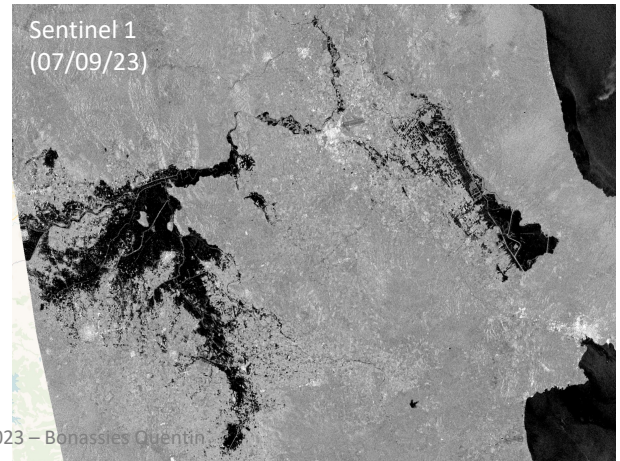
SAR image
Before event

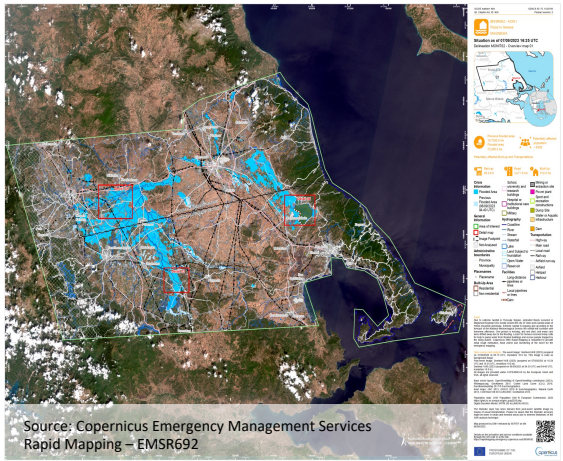




Aerial photos
During event

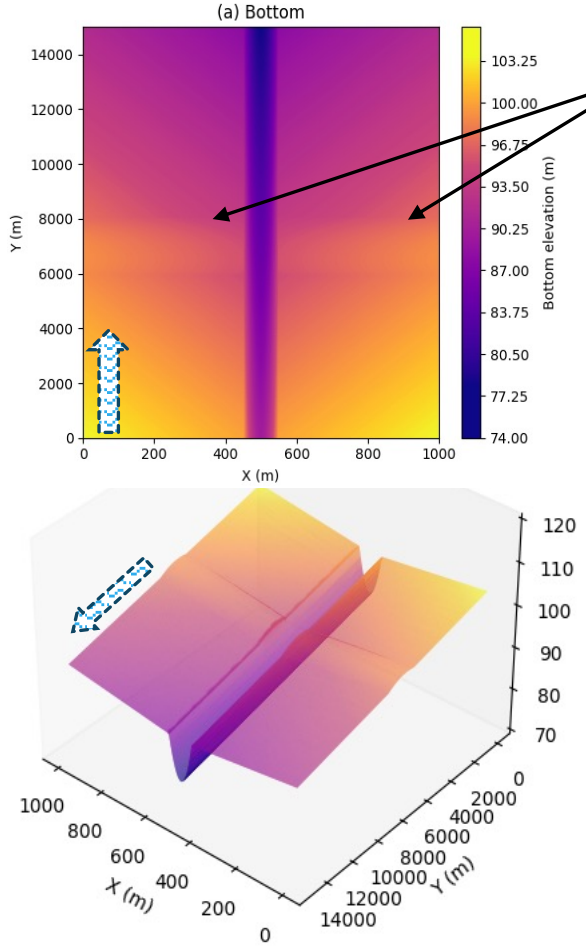
SAR image
During event





Copernicus
Emergency
Services
Rapid Mapping

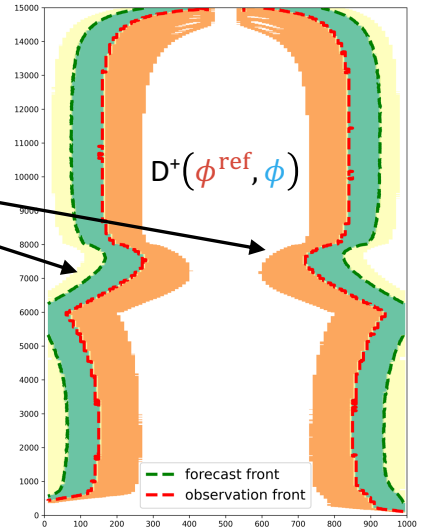
The Valley test case

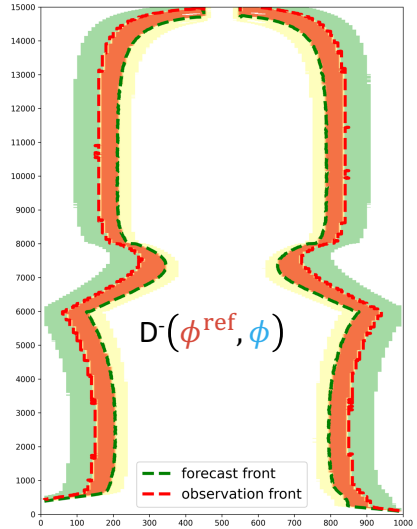


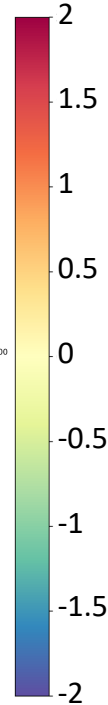
- Two extrusions in floodplains
- Constant slope in x-axis: 0.8%
- Uniform slope in floodplains: 8%

- Constant water flowrate
- 12 hours simulated
- Running time: around 2 min on 6 nodes of Intel Xeon Gold 6140 core (CERFACS HPC)

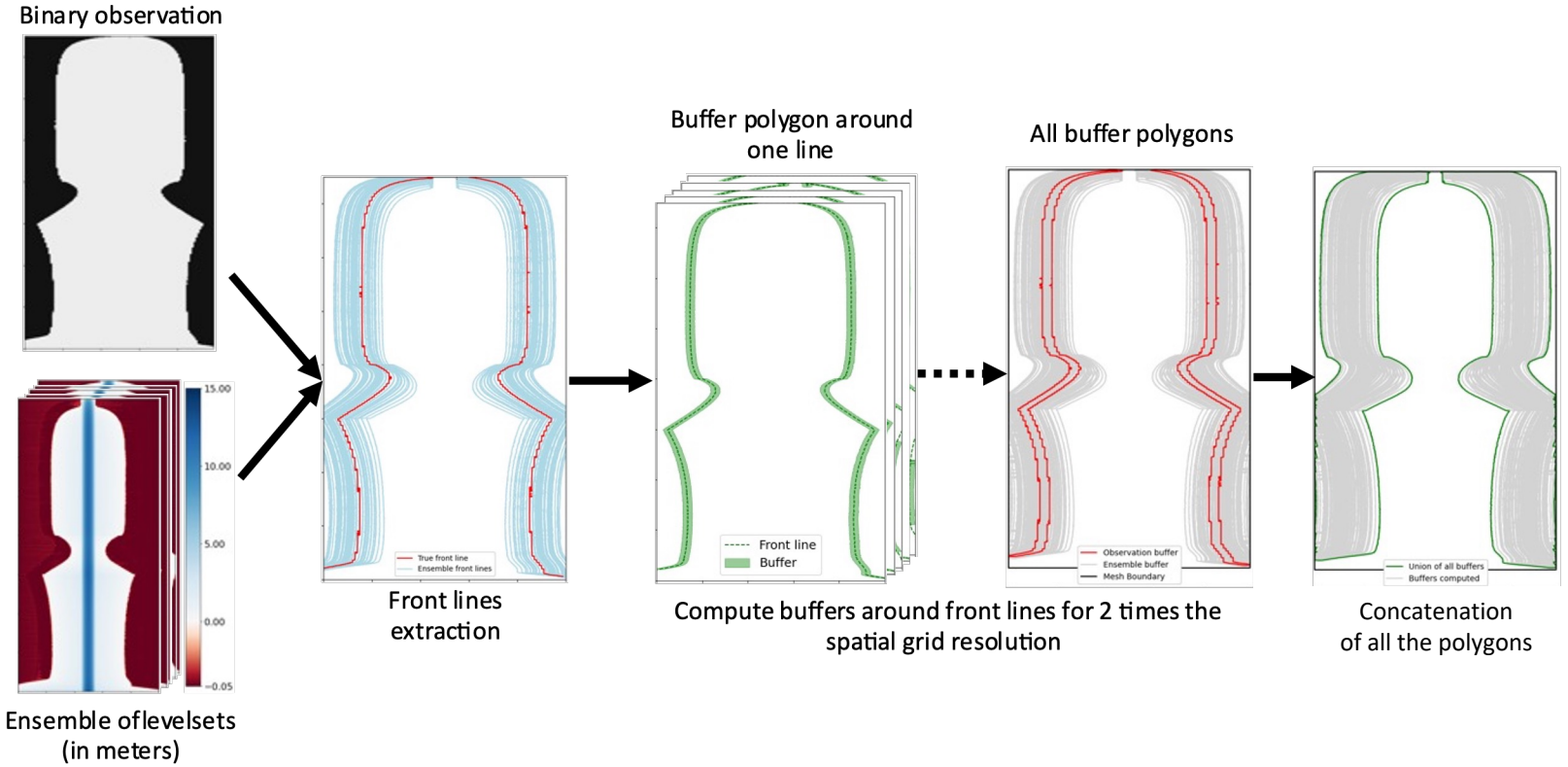
- Chan-Vese distance computed between an ensemble of forecasted fronts and a reference
- The reference, ϕ^{ref}, can be
  - an observed front,
  - or the ensemble mean front







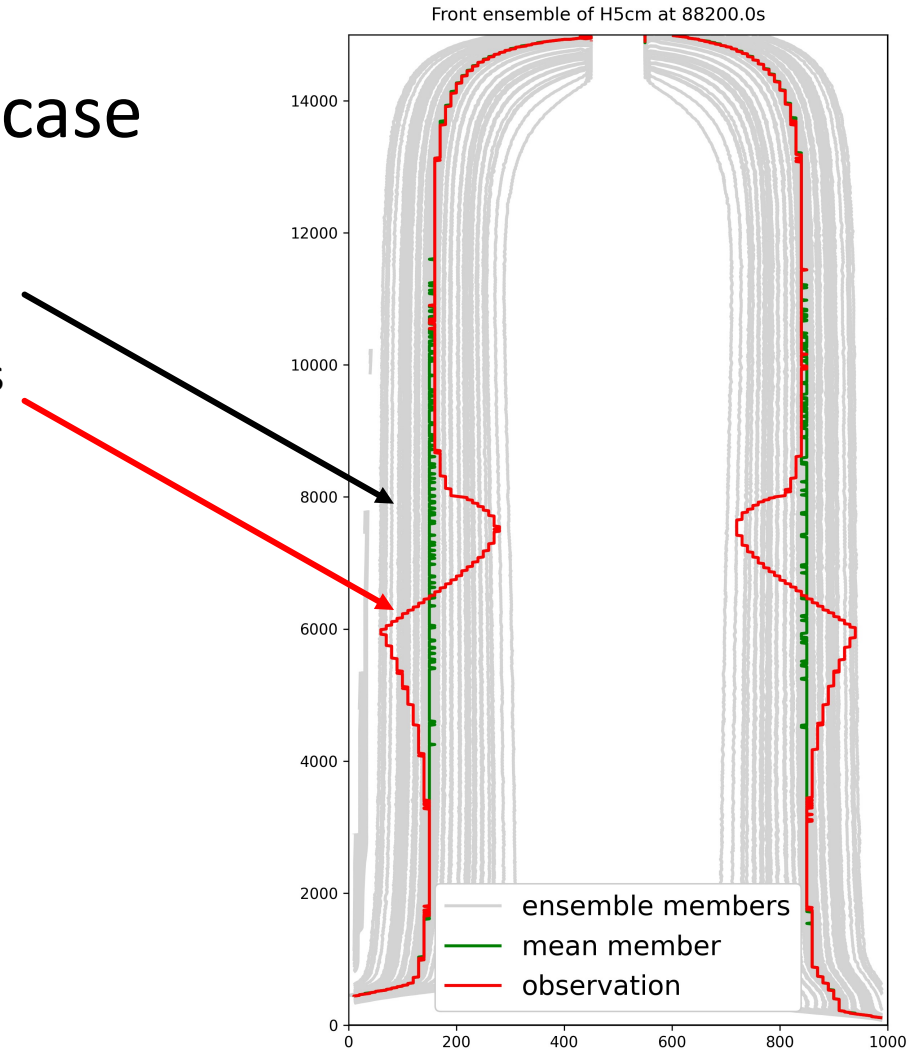
Selection of pixel around the wet/dry interface



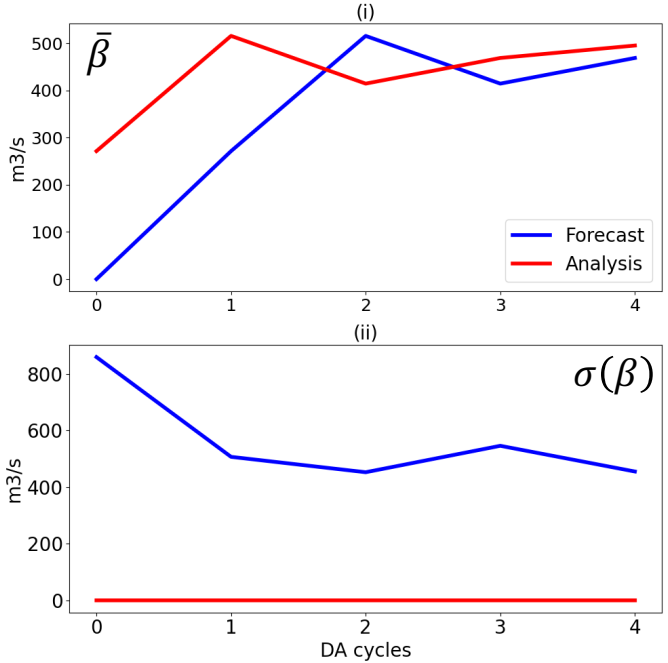
Changes in the Valley test case

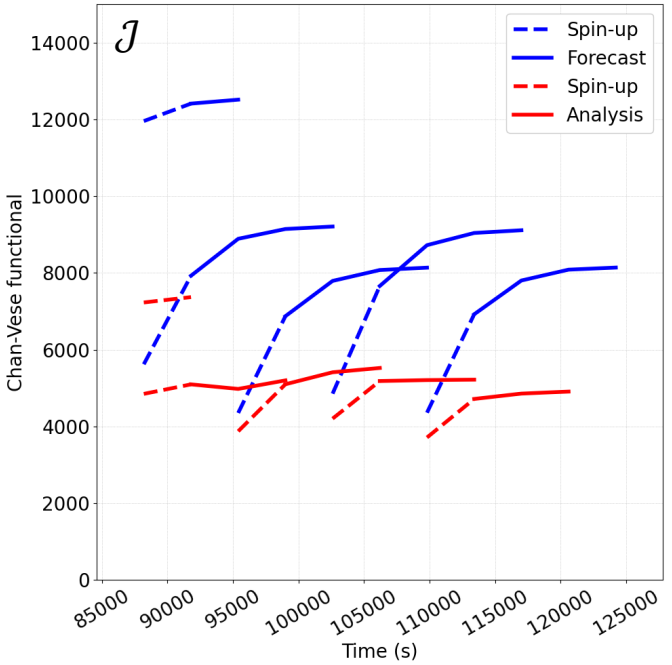
- Forecast: floodplains without extrusions
- Observations: floodplains with extrusions
- $Q_{up,\ truth} = 5000\ m^3/s$

$Q_{up,\ prior}\ (m^3/s)$	4500	5000	5500



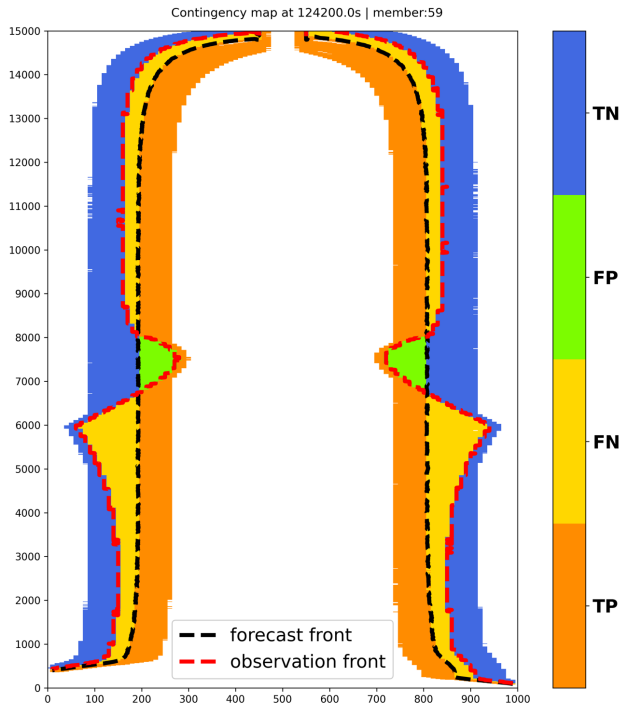
$Q_{up,\ prior} = 4500\ m^3/s$



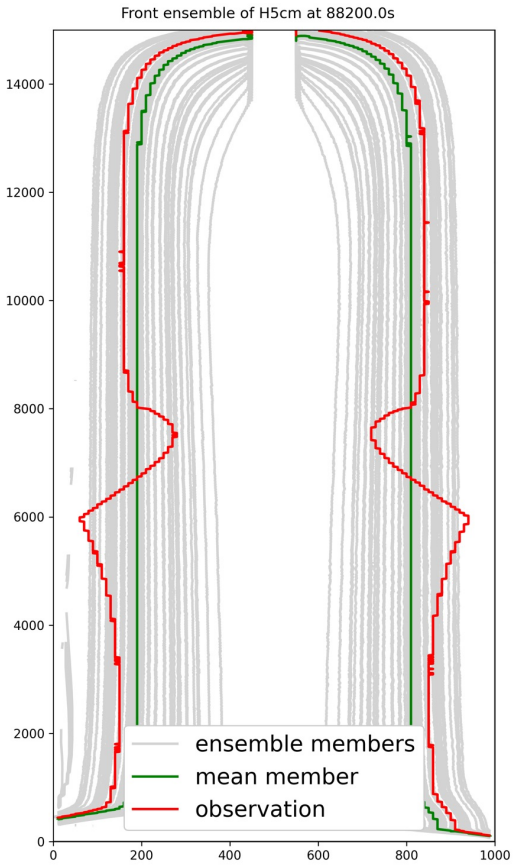


$Q_{up,\ prior} = 4500\ m^3/s$

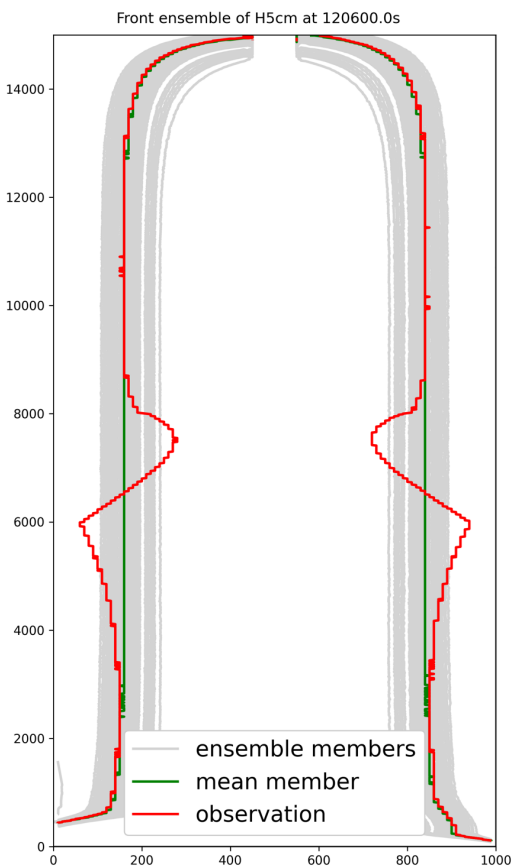
Contingency map example



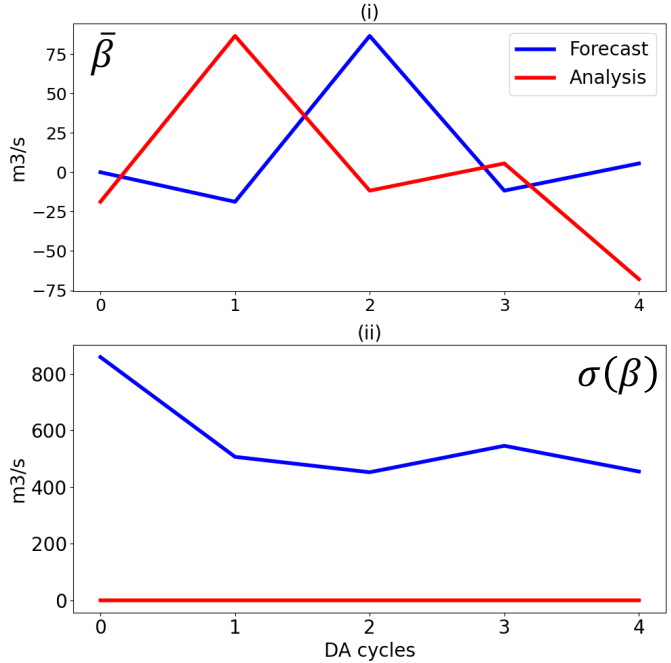
Forecast@88200s

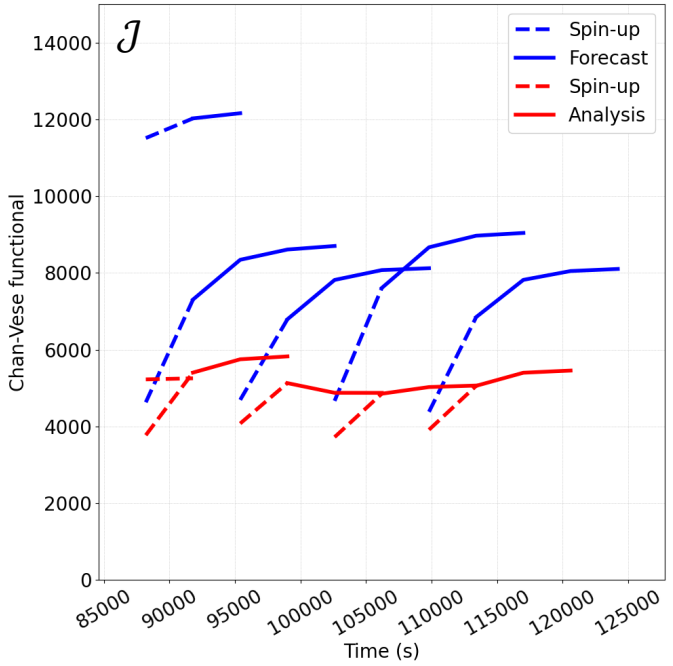


Forecast@120600s



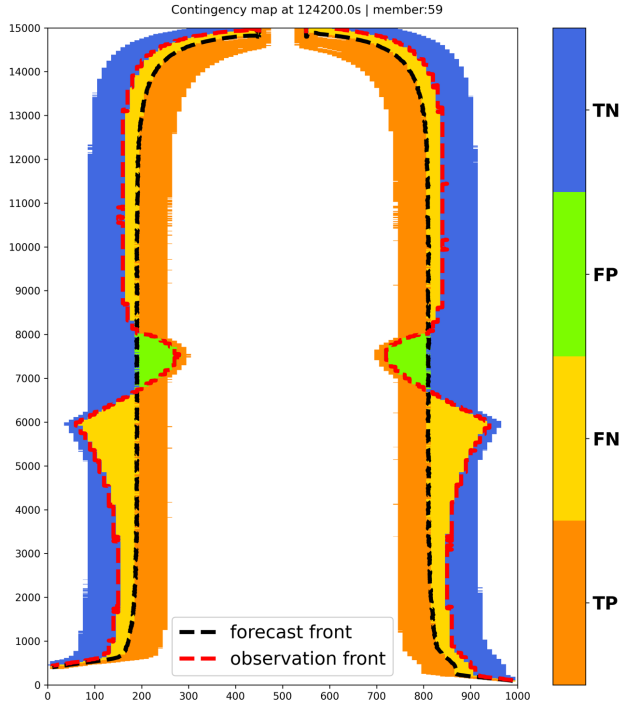
$Q_{up,\ prior} = 5000\ m^3/s$



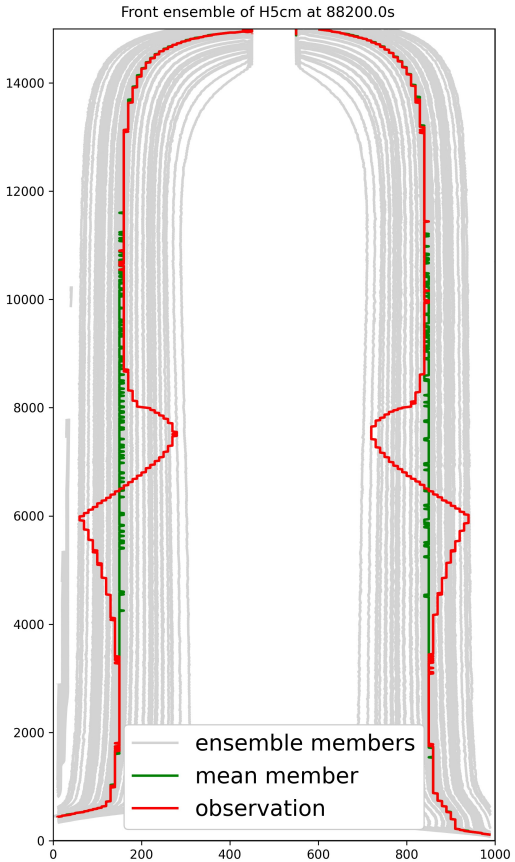


$Q_{up,\ prior} = 5000\ m^3/s$

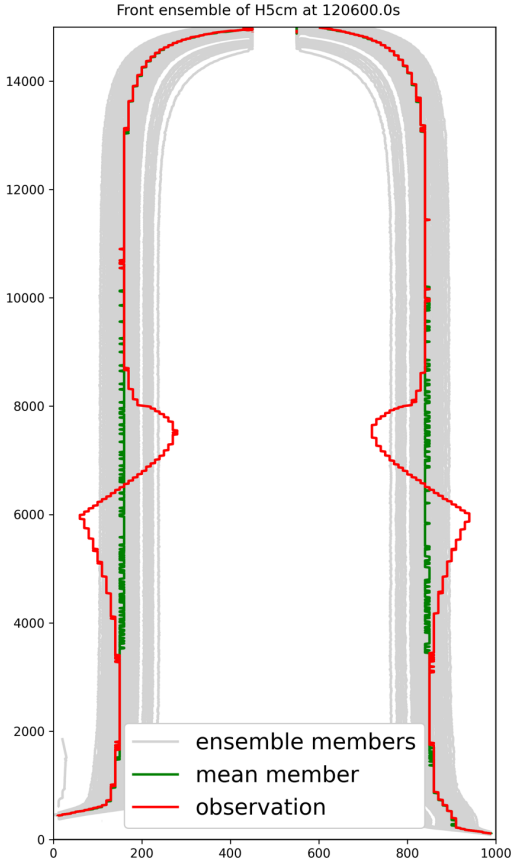
Contingency map example



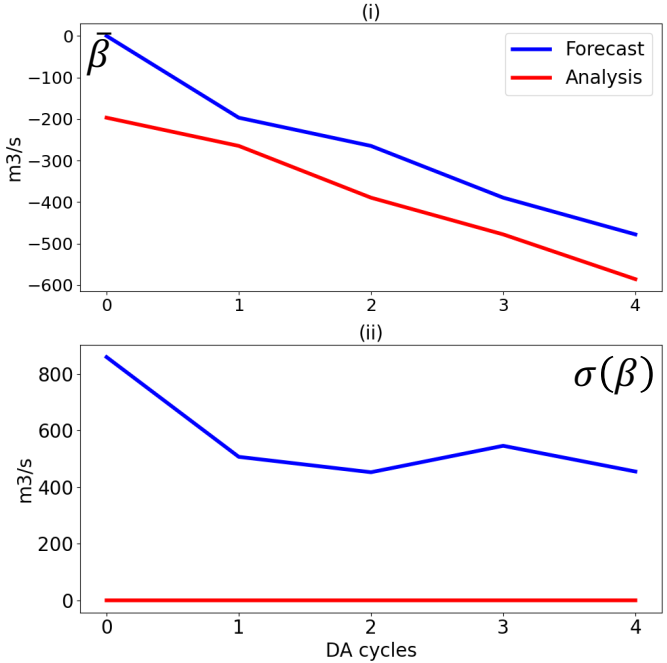
Forecast@88200s

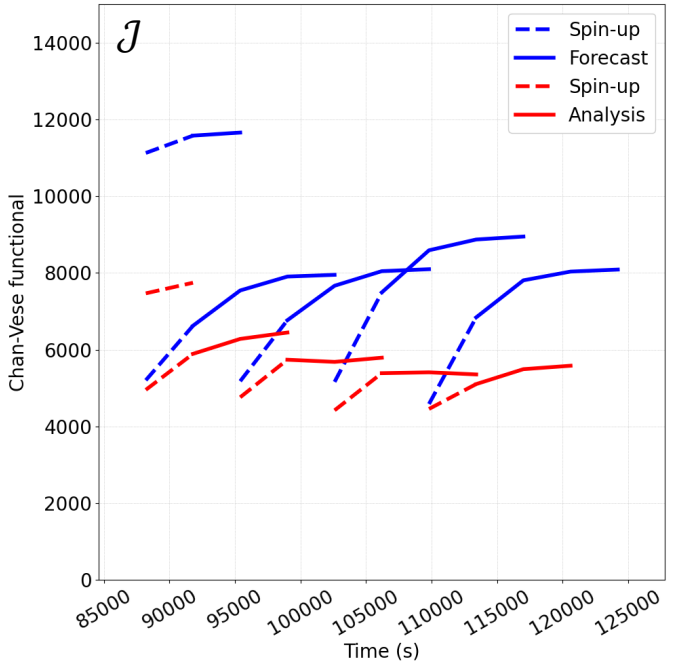


Forecast@120600s



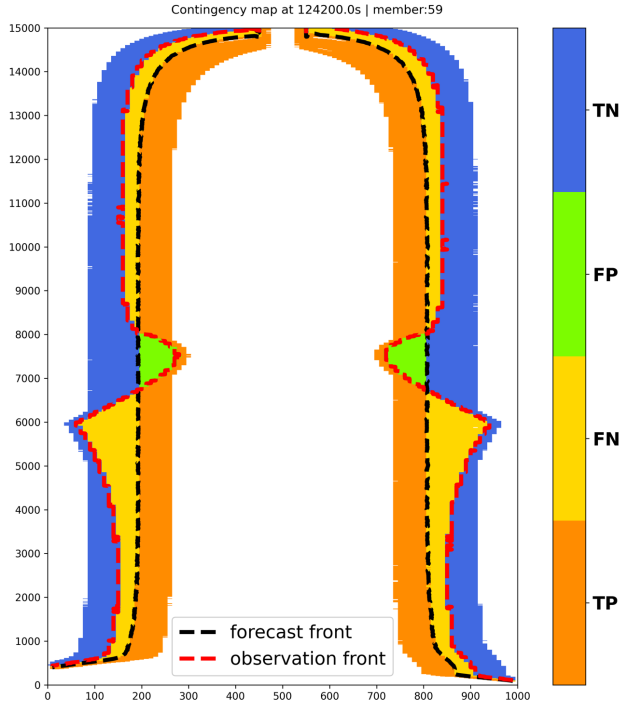
$Q_{up,\ prior} = 5500\ m^3/s$



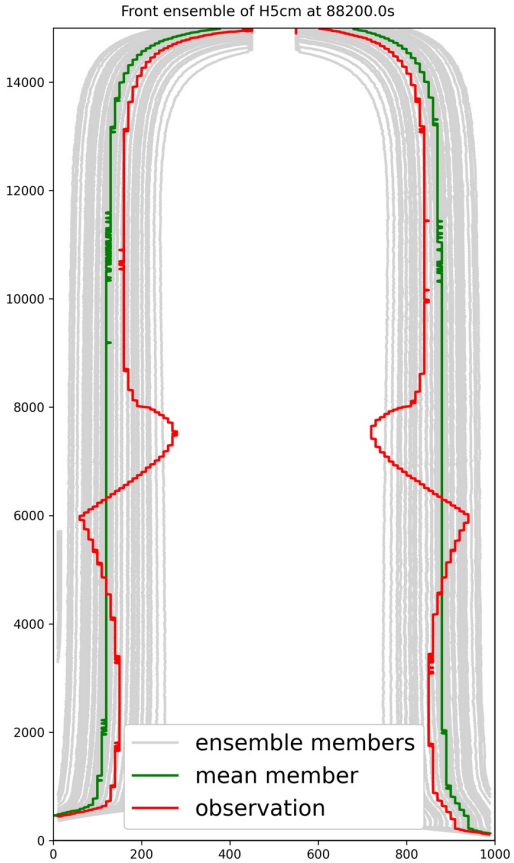


$Q_{up,\ prior} = 5500\ m^3/s$

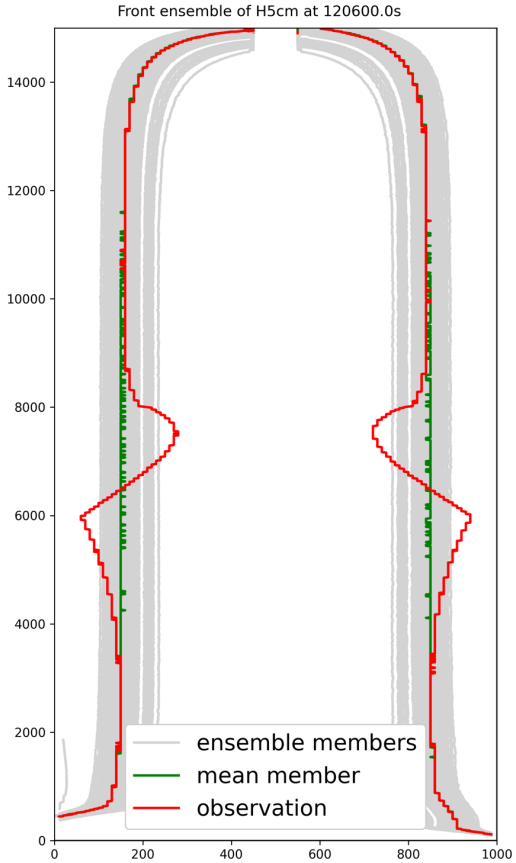
Contingency map example



Forecast@88200s



Forecast@120600s

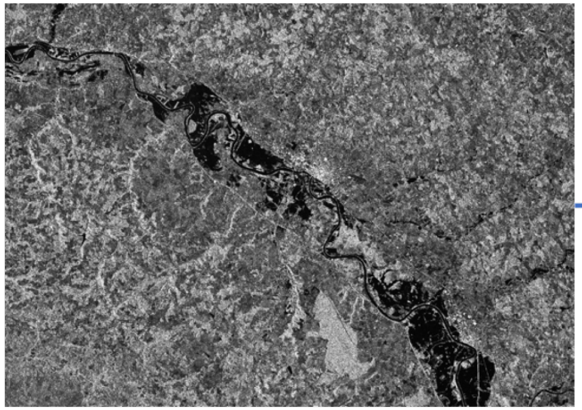


Data Assimilation of RS images

SAR-derived flood extent computation: FloodML Algorithms

Recent flood event on the studied catchment in January-February 2021

Sentinel-1 SAR image

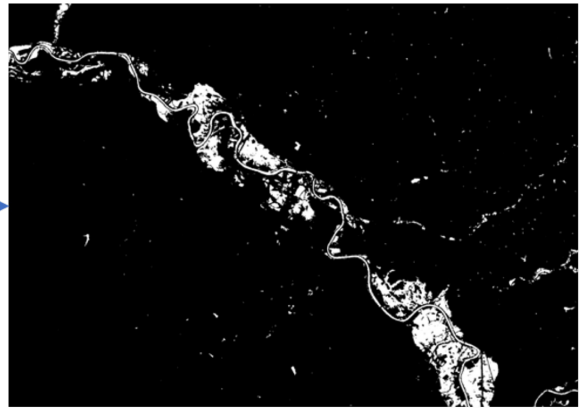




FloodML algorithms

Training database

Derived flood map (binary wet/dry)



Ground sampling distance:
10 x 10 meters

Feature	In-situ observations	Satellite observations
Time	Every 5 or 15 minutes	12 obs throughout the event
Location	3 predetermined locations	Cover the whole catchment

Source:

FloodML: Ketting *et al. (2021);*

Random Forest: Peña-Luque *et al.* (2021)